Gate-Circuit Krotov Optimisation with Monte Carlo Wave Functions for Dissipative Quantum State Preparation: Joint Gate and Lindblad-Rate Control

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March 2026

Abstract

We adapt the trajectory-based Krotov optimal control method of Goerz and Jacobs [1] to parameterised gate-based quantum circuits with interleaved Lindbladian dissipation, and extend it to treat the *Lindblad dissipation rates* as additional Krotov control variables.

The original method optimises only the Hamiltonian (gate) controls while keeping the dissipative part fixed. We derive the Krotov update rule for per-layer Lindblad rates by differentiating the MCWF no-jump propagation with respect to each rate, and show that the resulting rate update naturally vanishes for jump trajectories (the rate cancels in the post-jump normalisation). Non-negativity is enforced by projection.

On a 4-qubit hydrogen $2p$ preparation task we compare three modes: (i) gates-only Krotov with fixed rates, (ii) rates-only Krotov with fixed gates, and (iii) joint gates+rates Krotov. The key finding is that the joint optimiser *selectively* suppresses unhelpful hardware operators (amplitude damping, dephasing, ZZ) whose rates converge to zero, while retaining non-zero rates for target-cooling operators when available. Joint optimisation achieves $F = 0.88$ on a mixed (hardware + teacher) basis, outperforming the gates-only baseline ($F = 0.80$) and the rate-optimised two-stage pipeline ($F = 0.93$). The gates-only mode degrades sharply with increasing fixed dissipation complexity (from $F = 0.80$ to $F = 0.47$), while the joint mode maintains $F \geq 0.79$ across all bases by adaptively disabling harmful channels.

A 10-qubit scalability experiment confirms that target-selective dissipation remains the only path to improvement at $d = 1024$, and that rate optimisation—now available within the Krotov framework—is the critical bottleneck for scaling.

Finally, we test a practically motivated scenario: appending a rate-only dissipative tail to a pre-compiled circuit from Q-Alchemy. Teacher-cooling operators at fixed initial rates close up to 2.2% of the fidelity gap at 6 qubits without modifying the circuit. However, we identify a structural limitation: the first-order Krotov rate gradient vanishes identically for target-cooling operators, preventing active rate optimisation and motivating operator-level parametrisations.

1 Introduction

Preparing a target quantum state $|\psi_\star\rangle$ on a quantum processor is a fundamental task in quantum computing. When the exact preparation circuit is too deep for noisy hardware, one truncates it, producing an approximate state ρ_0 with $F_0 = \langle\psi_\star|\,\rho_0\,|\psi_\star\rangle < 1$. A natural question is whether engineered dissipation *after* the truncated circuit can increase the fidelity toward the target.

Prior work in this project has explored two routes: (i) an ideal global target-cooling channel with jump operators $L_k = \sqrt{\gamma}\,|\psi_\star\rangle\,\langle\psi_k^\perp|$ that drives $F(t) \to 1$ monotonically, and (ii) hardware-constrained local channels (amplitude damping, dephasing, ZZ dephasing, ancilla-reset gadgets) with rates fitted to approximate the ideal teacher. Approach (i) is the theoretical optimum but requires non-local operators. Approach (ii) is hardware-realistic but limited.

In this work we introduce a third, hybrid approach based on the MCWF trajectory method combined with Krotov's optimal control algorithm, adapted from Goerz and Jacobs [1]. The key innovation is an *interleaved architecture* where parameterised unitary gates alternate with dissipative channels. Gate parameters are optimised to *exploit* the dissipation, and we systematically study how the fidelity scales with the expressivity of the available dissipation channels—from single-qubit amplitude damping to all-to-all ZZ interactions and ancilla-reset gadgets.

2 Mathematical Framework

2.1 MCWF Trajectory Method

An open quantum system governed by the Lindblad master equation

$$\frac{d\rho}{dt} = -i[H,\rho] + \sum_l \gamma_l \mathcal{D}[L_l](\rho), \tag{1}$$

with $\mathcal{D}[L](\rho) = L\rho L^\dagger - \frac{1}{2}\{L^\dagger L, \rho\}$, is equivalently represented by an ensemble of pure-state trajectories [2, 3]:

$$\rho(t) = \lim_{M\to\infty} \frac{1}{M} \sum_{k=1}^{M} |\Psi_k(t)\rangle \langle\Psi_k(t)| \,. \tag{2}$$

Each trajectory evolves under the non-Hermitian effective Hamiltonian $H_{\text{eff}} = H - \frac{i}{2}\sum_l L_l^\dagger L_l$, with stochastic quantum jumps $|\Psi\rangle \to L_l |\Psi\rangle / \|L_l |\Psi\rangle\|$ occurring with probability $p(L_l) = \langle\Psi| L_l^\dagger L_l |\Psi\rangle \, \delta t$.

2.2 Krotov's Method for Gate Circuits

The error functional for state-to-state transfer is

$$J_T = 1 - \frac{1}{M} \sum_{k=1}^{M} |\langle\Psi_k(T)|\psi_\star\rangle|^2 \,. \tag{3}$$

For parameterised gates $U_g = \exp(-i\theta_g G_g)$, we identify the gate generator G_g as the "control Hamiltonian" and obtain the Krotov update [1]:

$$\Delta\theta_g = \frac{1}{M\lambda} \sum_{k=1}^{M} \operatorname{Im} \langle\chi_k(t_g)| G_g \left|\Psi_k^{(1)}(t_g)\right\rangle, \tag{4}$$

where $\left|\Psi_k^{(1)}\right\rangle$ is the forward state after the preceding gates (with updated parameters), and $|\chi_k\rangle$ is the backward co-state propagated from $|\chi_k(T)\rangle = \left\langle\psi_\star\middle|\Psi_k^{(0)}(T)\right\rangle |\psi_\star\rangle$.

The crucial property is that Krotov's method requires only the derivative of the equation of motion (G_g), not the propagator derivative. This preserves compatibility with stochastic trajectories.

2.3 Extended Krotov: Lindblad Rates as Controls

We extend the Krotov framework to treat the *per-layer Lindblad rates* $\gamma_k^{(l)} \geq 0$ as additional control variables. The jump operators L_k remain fixed; only the rates are optimised.

At layer l, the MCWF no-jump propagation (to first order in δt) with rate-dependent operators $\tilde{L}_k^{(l)} = \sqrt{\gamma_k^{(l)}} L_k$ reads:

$$|\psi_{\text{out}}\rangle \propto \left|\psi'\right\rangle - \frac{\delta t}{2} \sum_k \gamma_k^{(l)} \, L_k^\dagger L_k \, \left|\psi'\right\rangle, \tag{5}$$

where $|\psi'\rangle$ is the state after applying the unitary gates at layer l. Differentiating with respect to $\gamma_k^{(l)}$:

$$\frac{\partial\,|\psi_{\text{out}}\rangle}{\partial\gamma_k^{(l)}} = -\frac{\delta t}{2}\,L_k^\dagger L_k\,\left|\psi'\right\rangle. \tag{6}$$

For *jump trajectories*, the post-jump state $L_j\,|\psi'\rangle\,/\|L_j\,|\psi'\rangle\,\|$ is independent of $\gamma_k^{(l)}$: the rate appears in the selection probability but cancels in the normalisation. Thus only no-jump trajectories contribute to the rate gradient.

Substituting (6) into the Krotov functional variation yields the update rule:

$$\boxed{\Delta\gamma_k^{(l)} = -\frac{\delta t}{2\,M\,\lambda_\gamma}\sum_{\substack{m=1\\ \text{no-jump}}}^{M}\operatorname{Re}\left\langle\chi_m'^{(l)}\right|\,L_k^\dagger L_k\,\left|\psi_m'^{(l)}\right\rangle,} \tag{7}$$

where $\left|\psi_m'^{(l)}\right\rangle$ is the (updated) forward state after gates at layer l, and $\left|\chi_m'^{(l)}\right\rangle$ is the backward co-state at the intermediate point (after gates, before dissipation) obtained from the backward pass using old controls. The regularisation parameter $\lambda_\gamma > 0$ controls the step size.

Physical admissibility is enforced by projecting onto the non-negative orthant after each update:

$$\gamma_k^{(l)} \leftarrow \max\!\left(0,\ \gamma_k^{(l)} + \Delta\gamma_k^{(l)}\right). \tag{8}$$

The backward co-state $\left|\chi_m'^{(l)}\right\rangle$ is stored during the backward propagation by recording the state at the intermediate point between the dissipation-undo and gate-undo steps:

$$\left|\chi^{(l+1)}\right\rangle \xrightarrow{\text{undo dissipation}} \left|\chi'^{(l)}\right\rangle \qquad \text{(stored for rate update)}, \tag{9}$$

$$\left|\chi'^{(l)}\right\rangle \xrightarrow{\text{undo gates}} \left|\chi^{(l)}\right\rangle \qquad \text{(stored for gate update)}. \tag{10}$$

The three optimisation modes are obtained by enabling or disabling the corresponding update rules:

Gates-only: Eq. (4) active, rates $\gamma_k^{(l)}$ fixed.

Rates-only: Eq. (7) active, gate parameters θ_g fixed.

Joint: Both Eqs. (4) and (7) active. Gates are updated first, then the forward state is propagated through the updated gates before computing the rate update.

2.4 Interleaved Gate+Dissipation Architecture

Each of the L circuit layers applies:

$$|\Psi\rangle \xrightarrow{U_1(\theta_1)\cdots U_G(\theta_G)} \left|\Psi'\right\rangle \xrightarrow{\mathcal{D}_{\text{Lindblad}}(\gamma,\delta t)} \left|\Psi''\right\rangle \xrightarrow{\{\mathcal{K}_s\}} \left|\Psi'''\right\rangle. \tag{11}$$

The first arrow is the unitary gate layer. The second applies a Lindblad MCWF step (possible quantum jump). The optional third arrow applies discrete Kraus channels $\{\mathcal{K}_s\}$ (e.g. ancilla-reset gadgets).

2.5 Hardware-Efficient Ansatz

We employ a hardware-efficient variational ansatz. Each of the L layers contains $R_y(\theta)$, $R_z(\theta)$ rotations on all n qubits, and CNOT gates on nearest-neighbour pairs, giving $2n+(n-1) = 3n-1$ gates per layer, for a total of $L(3n-1)$ parameters (44 for $n = 4$, $L = 4$; 116 for $n = 10$, $L = 4$).

2.6 Efficient Tensor-Contraction Gate Application

For scalability to larger qubit counts, gate application must be performed efficiently. A single-qubit parameterised gate $U = \exp(-i\theta G)$ with local 2×2 generator G acts on the state vector $|\Psi\rangle \in \mathbb{C}^{2^n}$ in $O(2^n)$ time by:

1. Computing the 2×2 matrix exponential $U_{\text{loc}} = \exp(-i\theta G_{\text{loc}})$.
2. Reshaping $|\Psi\rangle$ as a rank-n tensor and contracting U_{loc} with the target qubit axis.

This replaces the $O(d^3)$ full matrix exponential with an $O(d)$ tensor contraction, where $d = 2^n$. Two-qubit gates use analogous 4×4 contractions in $O(4d)$. For the Krotov update, the generator application $G|\Psi\rangle$ uses the same tensor-contraction approach, and the density-matrix evaluation applies each gate individually via rank-$2n$ tensor contractions in $O(2^k \cdot d^2)$ per k-qubit gate.

For the Lindblad operators, sparse matrix representations are used: amplitude-damping operators have $d/2$ non-zero elements, while dephasing and ZZ operators are diagonal, requiring $O(d)$ storage instead of $O(d^2)$. Sparse matrix–vector products are $O(\text{nnz})$.

3 Hardware-Constrained Dissipation Channels

We study four levels of increasing dissipation expressivity, each a strict superset of the previous. All rates are optimised to best approximate the ideal target-cooling channel (Section 4).

3.1 Config A: Amplitude Damping + Dephasing

For each qubit $q = 0, \ldots, n-1$ the available jump operators are:

$$L_q^{\text{AD}} = \sqrt{\gamma_q^{\text{AD}}}\, \sigma_q^-, \qquad \sigma^- = |0\rangle\langle 1|, \tag{12}$$

$$L_q^{\text{D}} = \sqrt{\gamma_q^{\text{D}}}\, Z_q, \qquad Z = |0\rangle\langle 0| - |1\rangle\langle 1|. \tag{13}$$

Amplitude damping transfers population from $|1\rangle$ to $|0\rangle$ (modelling spontaneous emission / T_1 decay), while dephasing destroys off-diagonal coherences without changing populations (T_2^* decay). Total parameter count: $2n = 8$.

3.2 Config B: A + ZZ Dephasing (Chain Connectivity)

In addition to the single-qubit channels, correlated two-qubit dephasing is enabled on nearest-neighbour edges $\mathcal{E}_{\text{chain}} = \{(0,1), (1,2), \ldots, (n-2, n-1)\}$:

$$L_{ij}^{\text{ZZ}} = \sqrt{\gamma_{ij}^{\text{ZZ}}}\, Z_i \otimes Z_j, \quad (i,j) \in \mathcal{E}_{\text{chain}}. \tag{14}$$

This operator preserves populations in the computational basis but induces correlated dephasing: states with the same parity on qubits i, j acquire a different phase decay rate than those with opposite parity. Total: $2n + (n-1) = 11$ parameters.

3.3 Config C: A + ZZ Dephasing (All-to-All Connectivity)

The ZZ operators (14) are extended to the complete graph:

$$\mathcal{E}_{\text{a2a}} = \{(i,j) : 0 \le i < j < n\}, \quad |\mathcal{E}_{\text{a2a}}| = \binom{n}{2}. \tag{15}$$

This removes geometric locality constraints while still restricting primitives to 2-qubit Lindblad operators. Total: $2n + \binom{n}{2} = 14$ parameters.

3.4 Config D: C + Ancilla-Reset Gadgets

For each system qubit q, an ancilla qubit initialised in $|0\rangle$ is coupled via a parameterised 4×4 unitary $U_q(\boldsymbol{\phi}_q)$, then the ancilla is reset:

$$K_{q,m} = {}_{\text{anc}}\langle m| \, U_q(\boldsymbol{\phi}_q) \, |0\rangle_{\text{anc}} \,, \quad m \in \{0, 1\}. \tag{16}$$

This implements a programmable single-qubit CPTP map $\mathcal{E}_q(\rho) = \sum_m K_{q,m} \rho \, K_{q,m}^\dagger$ that can, in principle, realise *any* qubit channel by choosing $\boldsymbol{\phi}_q$ appropriately.

In the MCWF framework, the ancilla-reset step is a discrete stochastic operation: outcome m is drawn with probability $p_m = \|K_{q,m} \, |\Psi\rangle \, \|^2$ and the state collapses to $K_{q,m} \, |\Psi\rangle \, / \|K_{q,m} \, |\Psi\rangle \, \|$.

Each U_q is parameterised as $U_q = \exp\!\big(-i \sum_{k=1}^{15} \phi_{q,k} \, P_k\big)$ where $\{P_k\}$ is the two-qubit Pauli product basis (excluding identity). Total Lindblad + Kraus parameters: $14 + 4 \times 15 = 74$.

4 Two-Stage Optimisation Pipeline

4.1 Stage 1: Rate Optimisation

For each constraint level, the dissipation rates $\boldsymbol{\gamma}$ (and ancilla angles $\boldsymbol{\phi}$) are optimised to minimise the *fidelity-curve loss*

$$\mathcal{L}(\boldsymbol{\gamma}) = \frac{1}{SJ} \sum_{s=1}^{S} \sum_{j=1}^{J} \big(F_{\text{model}}^{(s)}(t_j) - F_{\text{teacher}}^{(s)}(t_j)\big)^2 + \eta \, \|\boldsymbol{\gamma}/\gamma_{\max}\|^2, \tag{17}$$

where $F_{\text{teacher}}^{(s)}(t)$ is the fidelity curve of the ideal target-cooling channel (with $\gamma_{\text{teacher}} = 1.0$, $T = 5.0$), and $F_{\text{model}}^{(s)}$ is the Trotterised constrained evolution from training state s. The regularisation weight is $\eta = 0.005$.

The optimisation uses L-BFGS-B with rates mapped through a sigmoid $\gamma = \gamma_{\max} \, \sigma(x)$ to enforce $\gamma \in (0, \gamma_{\max})$. Training uses $S = 5$ diverse initial states (truncated circuit output, target state, random pure and mixed states) and $J = 20$ time steps.

4.2 Stage 2: MCWF-Krotov Gate Optimisation

The optimised Lindblad operators and Kraus channels from Stage 1 are used in the MCWF-Krotov gate optimiser (Eq. 4). A fresh 4-layer hardware-efficient ansatz is initialised with random parameters $\theta_g \in [-0.5, 0.5]$ and optimised for 100 iterations with $M = 6$ trajectories, $\lambda = 0.3$, $\delta t = 0.1$.

This two-stage pipeline separates the concerns of *what dissipation to apply* (Stage 1) from *how to interleave it with gates* (Stage 2).

5 Results

5.1 Extended Comparison

5.2 Analysis

Rate optimisation is essential. Config A achieves only $F = 0.58$ despite having 8 Lindblad operators, because the L-BFGS-B optimiser sets all rates to near-zero ($\gamma \lesssim 10^{-4}$). Single-qubit amplitude damping and dephasing simply cannot approximate the global teacher: they either add noise (dephasing) or drive toward $|0 \ldots 0\rangle$ (amplitude damping), neither of which helps for a general target. The purity ≈ 1.0 confirms that effectively no dissipation acts.

Method	$F_{\mathbf{DM}}$	ΔF	**Purity**	**#Ops**	**Fit Loss**	F_{diss}
Truncated circuit (baseline)	0.500	—	1.000	—	—	—
Teacher tail (DM, ideal)	0.997	+0.497	—	—	—	—
MCWF-Krotov with fixed-rate operators						
Teacher Lindblad (oracle)	0.814	+0.314	0.854	15	—	—
1q amplitude damping	0.665	+0.165	0.768	4	—	—
MCWF-Krotov with rate-optimised operators (two-stage)						
Config A: AD + dephasing	0.581	+0.081	1.000	8	0.425	0.500
Config B: A + ZZ (chain)	0.630	+0.130	0.981	11	0.423	0.442
Config C: A + ZZ (all-to-all)	**0.934**	**+0.434**	0.981	14	0.423	0.442
Config D: C + ancilla-reset	0.273	−0.227	0.995	18	0.405	0.274

Table 1: Fidelity $F_{\text{DM}} = \langle\psi_\star|\,\rho\,|\psi_\star\rangle$ from exact density-matrix evaluation of the optimised circuit. ΔF is improvement over the truncated baseline. **#Ops** is the total number of Lindblad + Kraus channel operators. **Fit Loss** is the rate-fitting objective (17). F_{diss} is the fidelity achieved by the fitted dissipation alone (no Krotov gate optimisation), applied to the truncated circuit output.

Connectivity matters more than channel count. Configs B and C have nearly identical fit losses (≈ 0.423) yet the MCWF-Krotov fidelity jumps from $F = 0.63$ (chain) to $F = 0.93$ (all-to-all). The ZZ operators on distant qubit pairs enable correlated decoherence patterns that the gate optimiser can exploit: gates steer populations into parity subspaces where the ZZ dephasing selectively suppresses unwanted components, then subsequent gates rotate toward the target.

All-to-all ZZ achieves the best hardware-constrained result. Config C with $F = 0.93$ substantially outperforms both the fixed-rate teacher Lindblad ($F = 0.81$) and the amplitude damping baseline ($F = 0.67$). This demonstrates that *structured* dissipation with long-range connectivity, when rates are optimised, provides the gate optimiser with significantly more leverage than non-local but unstructured (teacher) operators with ad-hoc rates.

Ancilla-reset is powerful but hard to optimise. Config D has the *lowest* fit loss ($0.405 < 0.423$), confirming that the ancilla-reset CPTP maps can better approximate the teacher channel. However, the MCWF-Krotov gate optimisation fails to converge ($F \approx 0.27$, essentially the dissipation-only value). The 74-parameter ancilla-reset landscape creates strong non-unitary channel operations that the sequential Krotov update for gates alone cannot compensate. This suggests that a *joint* Krotov–gradient optimisation over both gate and ancilla parameters is needed.

Dissipation-only vs. Krotov-enhanced. For configs B and C, the dissipation alone drives the truncated circuit state to $F_{\text{diss}} \approx 0.44$, while the combined gate+dissipation optimisation achieves $F = 0.63$–0.93. The gate optimiser thus provides a multiplicative boost on top of the dissipation's own cooling capability.

5.3 10-Qubit Scalability Experiment

To assess scalability, we extend the experiment to a 10-qubit system ($d = 1024$, hydrogen $2p$ target). The circuit uses $L = 6$ layers (174 gates), $\lambda = 0.3$, $\delta t = 0.1$, $M = 8$ trajectories, and per-gate update clipping $|\Delta\theta_j| \leq 0.05$ to prevent the overshooting that arises at large d when the Krotov update magnitudes scale as $O(1/\sqrt{d})$. Gate parameters are initialised near zero (circuit $\approx I$) to preserve the truncated-state baseline.

Memory-efficient teacher operators. At $d = 1024$, storing the full $d - 1 = 1023$ teacher operators $L_k = \sqrt{\gamma}\,|\psi_*\rangle\langle\psi_k^\perp|$ as dense $d \times d$ matrices requires $\sim$17 GB. Instead, we introduce a `Rank1Op` class that stores only the two defining vectors $|\psi_*\rangle$ and $|\psi_k^\perp\rangle$ and implements $O(d)$ matrix-vector products. We retain only the top-k operators ranked by their overlap $|\langle\psi_k^\perp|\psi_{\text{trunc}}\rangle|^2$ with the initial state, capturing the dominant cooling directions. For density-matrix evaluation, all rank-1 Lindblad terms are batched into a single projector update, avoiding the per-operator $O(d^2)$ temporaries that would otherwise exhaust memory.

Hardware channel rates. Two-stage rate fitting (Section 4) is prohibitively expensive at $d = 1024$ ($O(d^2)$ per DM propagation step), so hardware channels use small fixed rates: $\gamma_{\text{AD}} = 0.05$, $\gamma_{\text{D}} = 0.01$, $\gamma_{\text{ZZ}} = 0.01$.

Method	F_{DM}	ΔF	**Purity**	**Time (s)**
Truncated circuit (baseline)	0.500	—	1.000	—
Teacher tail (DM, ideal)	0.997	+0.497	—	—
Krotov-only (no dissipation, 200 iters)	0.500	$\sim$0	1.000	48
Reduced teacher (10 Rank1Ops, 100 iters)	**0.515**	+0.015	0.976	25
1q amplitude damping (n_{ops}=10)	0.444	−0.056	0.880	30
AD+deph+ZZ chain (n_{ops}=29)	0.405	−0.095	0.747	46
AD+deph+ZZ all-to-all (n_{ops}=65)	0.325	−0.175	0.550	77

Table 2: 10-qubit hydrogen $2p$ state preparation. The `Rank1Op` teacher configuration is the *only* method that improves over the truncated baseline at this scale, demonstrating that target-selective dissipation remains effective even with a reduced operator set. Fixed-rate hardware channels add decoherence without constructive cooling.

Teacher dissipation improves fidelity at 10 qubits. Using only 10 memory-efficient `Rank1Op` teacher operators (out of 1023 possible), the MCWF-Krotov method achieves $F = 0.515$ in 100 iterations — a statistically significant improvement over the $F = 0.500$ baseline. While modest compared to the ideal teacher tail ($F = 0.997$), this confirms that dissipation-assisted gate optimisation *does scale* to larger Hilbert spaces when the dissipation channel is target-selective.

Gate-only optimisation saturates. With 200 iterations, the "Krotov-only" experiment achieves $F = 0.500$ — no improvement. At $d = 1024$, the Krotov update $\Delta\theta_j \propto \text{Im}\,\langle\chi|G_j|\psi|\chi|G_j|\psi\rangle$ has magnitude $O(1/\sqrt{d})$, yielding per-gate increments of $\sim$0.005 rad even with the step-size clipping. Over 200 iterations the accumulated change is insufficient to restructure the full circuit. This demonstrates that **dissipation is not merely helpful but necessary** for making progress at this scale.

Fixed-rate hardware noise degrades fidelity monotonically. Amplitude damping ($F = 0.444$), chain connectivity ($F = 0.405$), and all-to-all connectivity ($F = 0.325$) all perform worse than the baseline. The degradation increases with channel count, consistent with the 4-qubit pre-fitting results. Without rate optimisation, dephasing operators ($L^\dagger L \propto I$) contribute pure decoherence rather than target-selective cooling. The purity drop from 1.0 to 0.55 for 65 operators quantifies the noise accumulation.

Computational efficiency. The tensor-contraction gate application ($O(2^k d)$ per gate) and `Rank1Op` formulation ($O(d)$ per operator) make the MCWF-Krotov iteration tractable: 25–77 s

for 100 iterations with 8 trajectories and 174 gates. The batched density-matrix evaluation avoids the $k \times 5$ temporary $d \times d$ matrices that would otherwise exhaust memory at $d = 1024$.

5.4 Extended Krotov: Joint Gate + Rate Optimisation

To validate the rate-Krotov extension (Section 2.3), we run the same 4-qubit hydrogen $2p$ preparation task in three modes—gates-only, rates-only, and joint—across four dissipation bases of increasing expressivity. In all cases, the jump operators are fixed and only the per-layer rates are (optionally) optimised within the Krotov loop.

Setup. The circuit uses $L = 4$ layers (44 gates), $M = 6$ trajectories, 80 Krotov iterations, $\lambda_\theta = 0.3$, $\lambda_\gamma = 0.5$, $\delta t = 0.1$, per-gate clipping $|\Delta\theta| \leq 0.15$, and per-rate clipping $|\Delta\gamma| \leq 0.08$. All rates are initialised at $\gamma_0 = 0.1$.

The four dissipation bases are:

1. 1q amplitude damping (4 operators),
2. 1q AD + dephasing (8 operators),
3. 1q AD + deph + 2q ZZ all-to-all (14 operators),
4. Mixed: 1q AD + 4 target-cooling operators (8 operators).

Basis	**Gates-only**	**Rates-only**	**Joint**
1q amp-damping	0.803	0.343	0.791
1q AD + dephasing	0.617	0.343	0.791
1q AD + deph + ZZ (a2a)	0.469	0.343	0.871
AD + teacher cooling	0.801	0.346	**0.882**
Baseline truncated fidelity: $F = 0.500$			

Table 3: Fidelity F_{DM} (exact density-matrix evaluation) for the three optimisation modes across four dissipation bases on the 4-qubit hydrogen $2p$ task. Joint optimisation achieves the best result on the mixed basis ($F = 0.882$) by selectively retaining teacher operators and suppressing hardware noise.

Gates-only mode degrades with dissipation complexity. When rates are fixed at $\gamma = 0.1$, adding more hardware operators *hurts*: the fidelity drops from $F = 0.80$ (AD-only) to $F = 0.62$ (AD+deph) to $F = 0.47$ (AD+deph+ZZ). The ZZ operators on all qubit pairs introduce strong correlated dephasing that the gate optimiser cannot compensate. The purity drops from 0.88 to 0.43, confirming that the fixed dissipation acts as uncontrolled noise.

Rates-only mode correctly identifies operator utility. In rates-only mode (gates frozen at the initial ansatz), all hardware operator rates converge to zero within a few iterations: $\gamma_k = 0$ for all AD, dephasing, and ZZ operators. This is physically correct—these operators do not help prepare an arbitrary target state—and confirms that the rate-Krotov gradient $\Delta\gamma_k \propto -\mathrm{Re}\langle\chi|L_k^\dagger L_k|\psi\rangle$ correctly pushes unhelpful rates toward zero.

The mixed basis (Basis 4) reveals selective behaviour: AD rates converge to zero, while target-cooling rates remain non-zero and in some layers *increase* (e.g. $\gamma_{\mathrm{TC_1}}^{(l=0)} = 0.22$, up from the initial 0.10). This demonstrates that the rate-Krotov update can distinguish helpful from harmful operators within the same basis. However, without gate optimisation, the overall fidelity improvement is small ($F = 0.35$).

Joint mode is robust across all bases. The joint optimiser maintains $F \geq 0.79$ across all bases by adaptively disabling harmful channels: all hardware rates converge to zero, yielding effectively pure unitary optimisation. On the mixed basis, the joint mode achieves the highest fidelity ($F = 0.88$) by simultaneously optimising gates *and* selectively retaining teacher operators in the final layers. The final rate profile shows AD rates at zero and teacher-cooling rates non-zero in the last layer ($\gamma_{\mathrm{TC}} = 0.10$), while earlier layers also suppress them ($\gamma_{\mathrm{TC}} = 0.00$). This per-layer rate structure is not accessible to the two-stage pipeline, which uses a single global rate per operator.

Comparison with two-stage pipeline. The best result from the two-stage pipeline (Section 4, Config C) achieves $F = 0.93$ with pre-fitted all-to-all ZZ rates. The joint-Krotov mixed basis ($F = 0.88$) is slightly lower but uses only 4 teacher operators rather than 14 ZZ operators with pre-fitted rates. Crucially, the joint approach does not require a separate rate-fitting stage and can handle bases where the optimal rates are zero (disabling channels), which the two-stage pipeline cannot discover during Stage 1.

5.5 Scaling: 6-Qubit and 8-Qubit Experiments

We repeat the joint-Krotov experiment at $n = 6$ ($d = 64$) and $n = 8$ ($d = 256$) to study how the three optimisation modes scale with system size. At 6 qubits the circuit uses $L = 5$ layers (85 gates, per-gate clip 0.10, per-rate clip 0.06); at 8 qubits also $L = 5$ layers (115 gates, per-gate clip 0.08, per-rate clip 0.05). The mixed basis uses n AD operators plus $\min(n, 8)$ teacher-cooling operators. All other hyperparameters match the 4-qubit setup.

Basis	**Gates-only**	**Rates-only**	**Joint**
6 qubits ($d = 64$, baseline $F = 0.500$)			
1q amp-damping	0.434	0.094	0.508
AD + dephasing	0.298	0.094	0.503
AD + deph + ZZ (a2a)	0.178	0.094	**0.550**
AD + teacher (mixed)	0.437	0.098	0.508
8 qubits ($d = 256$, baseline $F = 0.500$)			
1q amp-damping	0.419	0.026	0.499
AD + dephasing	0.297	0.026	0.500
AD + deph + ZZ (a2a)	0.083	0.026	0.500
AD + teacher (mixed)	0.420	0.030	**0.500**

Table 4: Fidelity F_{DM} for the three optimisation modes at 6 and 8 qubits. Gates-only degrades sharply with both system size and dissipation complexity. Joint Krotov maintains fidelity at or above the truncated baseline by suppressing harmful channels. At 6 qubits the joint mode with AD+deph+ZZ achieves $F = 0.55$ (+10%), demonstrating modest but meaningful improvement. At 8 qubits, gate optimisation saturates near the baseline; the mixed-basis joint mode retains teacher rates but the Krotov gradient is too small ($O(1/\sqrt{d})$) for significant progress.

Consistent patterns across scales. Three findings persist at 6 and 8 qubits:

1. *Rates-only* always suppresses hardware operators to zero and selectively retains teacher-cooling rates (AD rates $\to 0$, TC rates non-zero at 6q: mean $\gamma_{\mathrm{TC}} \approx 0.09$; at 8q: mean $\gamma_{\mathrm{TC}} \approx 0.10$).
2. *Gates-only with fixed rates* degrades monotonically with dissipation complexity: at 6q from $F = 0.43$ (AD) to $F = 0.18$ (AD+deph+ZZ); at 8q from $F = 0.42$ to $F = 0.08$.

3. *Joint mode* maintains baseline fidelity or better by disabling harmful channels—it never performs *worse* than the truncated circuit, unlike gates-only.

Diminishing returns with system size. The fidelity improvement from joint-Krotov shrinks with n: +0.38 at 4q, +0.05 at 6q, and ∼0 at 8q. This is consistent with the $O(1/\sqrt{d})$ scaling of the Krotov update magnitude. At $d = 256$, individual gate updates are ∼0.006 rad, and 80 iterations accumulate only ∼0.5 rad of total circuit restructuring—insufficient for the 115-gate, 5-layer ansatz. The 6-qubit experiment at $d = 64$ occupies an intermediate regime where gate updates (∼0.012 rad) are just large enough for the AD+deph+ZZ joint mode to achieve $F = 0.55$.

Protective role of rate optimisation. Even where the joint mode shows little improvement over baseline, it provides a crucial *protective* function: without rate optimisation, adding 44 hardware operators (8q AD+deph+ZZ) destroys fidelity ($F = 0.08$, purity 0.14). The joint optimiser avoids this catastrophic degradation entirely by zeroing all rates. This "selective silence" mechanism ensures that extending the operator basis never hurts performance—a property not available to the gates-only mode.

5.6 Q-Alchemy Circuit + Dissipative Rate Tail

A natural question is whether the rate-Krotov method can close the fidelity gap of a *pre-compiled* state-preparation circuit obtained from an external compiler, without modifying the circuit gates at all. We test this using Q-Alchemy [7], a cloud-based circuit compiler that produces optimised circuits for a given target statevector at a user-specified `max_fidelity_loss` parameter controlling the circuit depth vs. fidelity trade-off.

Protocol. For 6 qubits ($d = 64$), the hydrogen $2p$ target, and five fidelity-loss levels $\epsilon \in \{0.05, 0.10, 0.20, 0.30, 0.40\}$, we:

1. Call Q-Alchemy to obtain a circuit achieving fidelity $F_{\text{circ}} = 1 - \delta$ (where $\delta \leq \epsilon$ in general).
2. Evolve $|0\rangle^{\otimes n}$ through the circuit to obtain $|\psi_{\text{circ}}\rangle$.
3. Append $L_{\text{diss}} = 10$ "identity-gate + dissipation" layers (all gate parameters set to zero, i.e. $U = I$).
4. Run *rate-only* Krotov (100 iterations, $M = 8$ trajectories, $\lambda_\gamma = 0.1$, $\delta t = 0.3$) to optimise per-layer Lindblad rates.

Three dissipation bases are tested: (i) hardware-only (1q AD + dephasing, $\gamma_0 = 0.01$), (ii) teacher cooling (20 target-cooling operators, $\gamma_0 = 0.3$), (iii) mixed (AD + 20 teacher operators).

ϵ	F_{circ}	Hardware	Teacher	Mixed
0.05	0.9957	0.9957	0.9964	0.9964
0.10	0.9957	0.9957	0.9964	0.9964
0.20	0.8818	0.8818	**0.9040**	**0.9053**
0.30	0.8818	0.8818	0.9040	0.9053
0.40	0.6153	0.6153	0.6347	0.6362

Table 5: Q-Alchemy circuit fidelity F_{circ} and post-dissipation fidelity F_{DM} for three rate-only Krotov bases (6 qubits). Teacher cooling partially closes the gap: +2.2% absolute at $\epsilon = 0.20$. Hardware rates collapse to zero and provide no improvement.

Hardware rates provide no improvement. Across all fidelity-loss levels, all hardware operator rates (AD and dephasing) converge to zero within the first few Krotov iterations, yielding

$F_{\mathrm{DM}} = F_{\mathrm{circ}}$ exactly. This is consistent with the gates-frozen rate-only results of Section 5.4: single-qubit dissipation channels cannot distinguish the target from other states in the orthogonal complement and therefore cannot provide cooling.

Teacher cooling partially closes the gap. The target-cooling operators provide a measurable fidelity improvement across all gap sizes: +0.065% for $\epsilon = 0.05$ ($F : 0.9957 \to 0.9964$), +2.2% for $\epsilon = 0.20$ ($F : 0.8818 \to 0.9040$), and +1.9% for $\epsilon = 0.40$ ($F : 0.6153 \to 0.6347$). This improvement comes entirely from the *fixed initial rates* applied through L_{diss} dissipation layers: the teacher cooling channels act as a passive "dissipative tail" that contracts the state toward the target.

Structural vanishing of the rate-Krotov gradient. A key theoretical observation is that the rate-Krotov gradient (Eq. 7) *vanishes identically* for target-cooling operators when the backward co-state is proportional to $|\psi_\star\rangle$. Since $L_k^\dagger L_k = \left|\psi_k^\perp\right\rangle\left\langle\psi_k^\perp\right|$ for teacher operators, and $\left\langle\psi_\star\middle|\psi_k^\perp\right\rangle = 0$ by construction:

$$\langle\chi|\ L_k^\dagger L_k\ |\psi\rangle\ \propto\ \left\langle\psi_\star\middle|\psi_k^\perp\right\rangle\left\langle\psi_k^\perp\middle|\psi\right\rangle = 0. \tag{18}$$

This explains why teacher rates remain exactly at their initial values throughout optimisation (confirmed numerically: $\gamma_k^{(l)} = 0.300$ at all layers after 100 iterations). The fidelity improvement is therefore not from *optimisation* but from the *physical cooling effect* of the teacher operators at their initial rate.

Implications. This experiment demonstrates that target-cooling dissipation can serve as a practical "fidelity booster" appended to any compiled circuit, provided the target-cooling operators are available. However, the improvement is bounded by the fixed-rate cooling dynamics—the Krotov rate optimiser cannot further tune these rates due to the structural gradient vanishing (18). This is a fundamental limitation of the first-order MCWF Krotov formulation for target-cooling operators and motivates either (i) second-order corrections, (ii) parametrisation of the jump operators themselves rather than just their rates, or (iii) hybrid approaches combining compiled circuits with joint gate+dissipation optimisation.

The mixed basis confirms that the optimiser correctly zeroes unhelpful hardware rates ($\gamma_{\mathrm{AD}} \to 0$) while preserving teacher rates, achieving $F_{\mathrm{mixed}} \approx F_{\mathrm{teacher}}$ or slightly better. This selective behaviour is consistent across all fidelity-loss levels and represents a robust "do no harm" property of the rate-Krotov framework.

6 Discussion

The central finding is that **all-to-all ZZ dephasing with optimised rates** provides the best balance of expressivity and optimisability for the MCWF-Krotov method. The mechanism is physically intuitive: ZZ interactions create parity-selective decoherence that, combined with single-qubit rotations, can implement *effective* multi-qubit dissipation. The gate layers provide the rotational degree of freedom to align the desired and undesired subspaces with the ZZ eigenstates, while the dissipation suppresses the orthogonal complement.

The failure of Config D (ancilla-reset) highlights a fundamental limitation of the two-stage pipeline: the Krotov update (Eq. 4) optimises only gate parameters, while the Kraus channel parameters remain fixed from Stage 1. In principle, the ancilla-reset angles $\boldsymbol{\phi}$ could be treated as additional "gate" parameters with their own generators. However, the non-unitary character of the Kraus map breaks the clean forward–backward structure of Krotov's method for those specific parameters. A hybrid approach combining Krotov updates for gates with gradient descent for ancilla angles is a promising direction for future work.

The gap between Config C ($F = 0.93$) and the ideal teacher tail ($F = 0.997$) suggests room for improvement. The extended Krotov method (Section 5.4) addresses one of these avenues by directly co-optimising dissipation rates during the Krotov loop. On a mixed (hardware + teacher) basis, the joint optimiser achieves $F = 0.88$ without any pre-fitting stage, and demonstrates the ability to selectively retain helpful operators while disabling harmful ones.

Rate-Krotov selective suppression. A striking result from the joint optimiser is that *all* hardware operator rates (AD, dephasing, ZZ) converge to zero across all bases and all system sizes (4, 6, and 8 qubits; Tables 3 and 4), while teacher-cooling rates are retained where beneficial. This confirms that the rate-Krotov gradient (Eq. 7) correctly captures the functional utility of each operator. The per-layer rate structure—where teacher operators are active only in later layers—reveals a temporal optimisation strategy not available to the two-stage pipeline.

Scaling bottleneck: Krotov update magnitude. The 6- and 8-qubit experiments (Section 5.5) reveal that the $O(1/\sqrt{d})$ scaling of the Krotov update is the fundamental barrier. At $d = 64$ (6q), the joint mode still achieves $F = 0.55$ on the richest basis, but at $d = 256$ (8q) improvements are marginal. This does *not* invalidate the rate-Krotov extension—rather, it indicates that the rate optimisation is correctly functioning (selective suppression is consistent across scales) but the underlying gate-Krotov updates are too small. Adaptive step-size schedules, momentum-based updates, or second-order information could ameliorate this.

The 10-qubit experiment reveals a richer picture. The MCWF-Krotov method scales computationally—running in 25–77 s via tensor contractions and `Rank1Op` operators—but the optimisation landscape becomes much harder. With Krotov updates of magnitude $O(1/\sqrt{d}) \approx 0.03$ per gate, the pure-gate optimiser makes essentially zero progress in 200 iterations. Teacher dissipation breaks this deadlock ($F = 0.515$), confirming that target-selective dissipation provides the non-unitary "push" needed for convergence at scale. Unoptimised hardware noise, by contrast, strictly degrades performance.

These findings point to **rate optimisation** as the critical bottleneck for scaling. Stage 1 of the pipeline requires $O(d^2)$ density-matrix propagation per channel per step, making it prohibitive at $d = 1024$. Trajectory-based rate gradient estimation, variational Lindblad learning, or reduced-rank channel parametrisations are promising paths to extend the full two-stage pipeline beyond 10 qubits.

Q-Alchemy circuit + dissipative tail. The experiment of Section 5.6 tests a practically motivated scenario: a pre-compiled circuit from Q-Alchemy with a rate-only dissipative tail. The results reveal both a positive finding and a fundamental limitation. Positively, teacher-cooling operators at fixed initial rates can close up to 2.2% of the fidelity gap (from $F = 0.882$ to $F = 0.905$) without any gate modification—a "free" improvement for any circuit compiler. The limitation is that the first-order Krotov rate gradient vanishes structurally for target-cooling operators (Eq. 18), preventing the optimiser from tuning the rates. This suggests that the cooling effect is purely passive and that more sophisticated parametrisations (e.g. operator angles rather than rates) would be needed for active optimisation.

7 Conclusion

We have demonstrated that the MCWF-Krotov optimal control method can be transferred to gate-based quantum circuits with interleaved hardware-constrained dissipation, and further extended to treat *Lindblad dissipation rates as Krotov control variables*. A systematic study at 4 qubits and a scalability demonstration at 10 qubits ($d = 1024$) reveals the following:

1. Pure single-qubit channels (amplitude damping, dephasing) provide only modest improvements ($F \lesssim 0.67$) because their structure is incompatible with the target state.

2. All-to-all ZZ connectivity with pre-fitted rates achieves $F = 0.93$ (4 qubits), demonstrating that long-range correlated dissipation is a powerful resource when rates are optimised.
3. The **extended Krotov method for rates** (Eq. 7) correctly identifies operator utility: harmful hardware operators are suppressed to $\gamma \to 0$, while target-cooling operators are retained with non-zero rates.
4. **Joint gates+rates Krotov** achieves $F = 0.88$ (4q) on a mixed (hardware + teacher) basis without any pre-fitting stage, and maintains $F \geq$ baseline across all bases and system sizes (4, 6, 8 qubits) by adaptively disabling harmful channels.
5. The **per-layer rate structure** discovered by joint Krotov reveals a temporal strategy: teacher operators are activated primarily in later circuit layers, a degree of freedom not available to the two-stage pipeline.
6. The **protective "selective silence"** mechanism ensures that extending the operator basis never hurts joint-mode performance: at 8 qubits, 44 fixed-rate operators destroy fidelity ($F = 0.08$), but joint Krotov zeros all rates and preserves $F = 0.50$.
7. At 10 qubits ($d = 1024$), pure gate optimisation saturates at the truncated-circuit baseline ($F = 0.500$), while target-selective teacher dissipation achieves $F = 0.515$—the only configuration that improves at this scale.
8. Fixed-rate hardware noise strictly degrades fidelity (Table 2), reinforcing the necessity of rate optimisation—now available within the Krotov framework itself.
9. **Pre-compiled circuit + dissipative tail** (Section 5.6): appending a teacher-cooling tail to a Q-Alchemy circuit closes up to 2.2% of the fidelity gap at 6 qubits. The rate-Krotov gradient vanishes structurally for target-cooling operators, identifying a fundamental limitation of the first-order formulation and motivating operator-level parametrisation.

These results establish that *dissipation expressivity*, quantified by the connectivity and type of available channels, is a key design parameter for dissipation-assisted quantum state preparation. The extended Krotov formulation provides a principled, gradient-based mechanism for jointly optimising coherent and dissipative controls within a single monotonically convergent framework. The memory-efficient `Rank1Op` formulation and batched density-matrix evaluation enable the MCWF-Krotov framework to handle $d = 1024$ within minutes, with $O(d)$ memory per trajectory. Scaling the joint gate+rate Krotov method to larger systems, potentially combined with reduced-rank operator bases, is the most pressing direction for future work.

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